Machine Learning Analysis Report

Project: Titanic Survival and Fare Prediction

1. Regression Analysis

The regression task focused on predicting passenger fares using Age, Pclass, and Sex. The primary metrics evaluated were Mean Absolute Error (MAE), Mean Squared Error (MSE), and the R2 Score.

	Model	Val MAE	Val MSE	Val R2	Test MAE	Test MSE	Test R2
0	Linear Regression	0.32	0.17	0.2996	0.30	0.15	0.3673
1	Decision Tree	0.22	0.16	0.3241	0.26	0.21	0.1590
2	Random Forest	0.24	0.14	0.4033	0.27	0.18	0.2624
3	Gradient Boosting	0.28	0.15	0.3939	0.29	0.16	0.3486
4	KNN	0.26	0.16	0.3204	0.27	0.17	0.2926
5	Voting Regressor	0.26	nan	nan	0.28	0.16	0.3443
6	Bayesian Ensemble	0.32	nan	nan	0.30	0.15	0.3679

Key Insight: Linear Regression emerged as the most robust model on the test set (R2: 0.3673), matching the Bayesian Ensemble. While complex models like Decision Trees showed lower error on validation, but they exhibited significant performance degradation on the test set, indicating overfitting. The high MSE compared to MAE suggests the presence of high-fare outliers that heavily penalize the squared error metrics.

2. Classification Analysis

The classification models aimed to predict passenger survival. Evaluation was based on Accuracy, Precision, Recall, and the harmonic F1-Score to ensure a balanced view of model performance.

	Model	Val Accuracy	Test Accuracy	Test Precision	Test Recall	Test F1-Score
0	Logistic Regression	75.70%	77.78%	0.72	0.78	0.75
1	Decision Tree	80.37%	75.93%	0.75	0.65	0.70
2	Random Forest	81.31%	75.93%	0.72	0.72	0.72
3	Gradient Boosting	78.50%	76.85%	0.74	0.70	0.72
4	KNN	77.57%	75.00%	0.70	0.72	0.71
5	SVC	75.70%	76.85%	0.76	0.67	0.71
6	Voting Ensemble (Best 3)	80.37%	76.85%	0.77	0.65	0.71

Key Insight: Logistic Regression provided the best generalization, achieving the highest Test Accuracy (77.78%) and F1-Score (0.75). The Voting Ensemble, while slightly less accurate overall, achieved the highest Test Precision (0.77), making it the most reliable model for avoiding False Positive survival predictions.

3. Conclusion

Across both tasks, simpler linear models demonstrated superior generalization to unseen data. The split was crucial in identifying that while ensemble methods and non-linear models performed exceptionally well during validation, they were more susceptible to noise in the Titanic dataset. For deployments, Logistic and Linear Regression offer the most stable foundations.